



**UNIVERSITI MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE AND MATHEMATICS**

**CSA 3013 MODELLING AND SIMULATION
LAB 6**

Prepared by: Ameer Muqrie Bin Abdul Mutalib (S72433)

Lecturer's Name

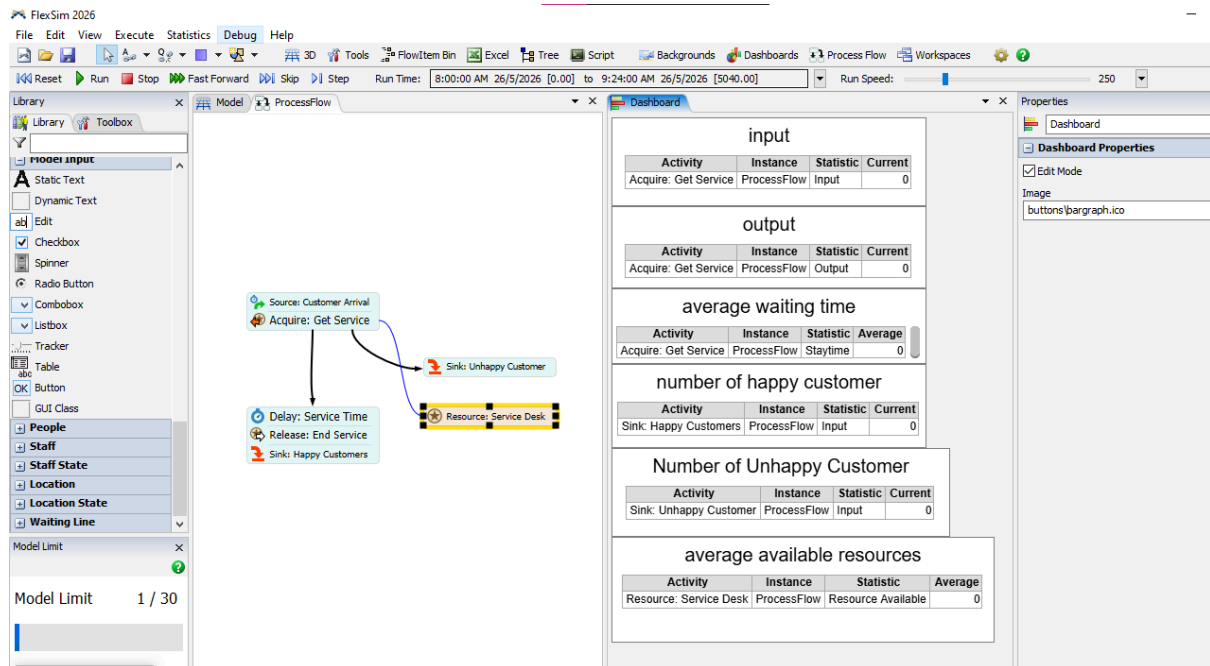
Dr. Najihah Binti Ibrahim

Demonstrator's Name

Madam Fatin Fatihah Yahya

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SCREENSHOT OF TASK



LINK FOR GDRIVE VIDEO

<https://drive.google.com/drive/folders/15F8jKsYcEah5ojPAURoubwuUpb3XwCMF?usp=sharing>

TASK 3 : EXPERIMENT AND RECORD RESULTS

1. Number Of Service Counters Experiment

Experiment	Resource Count	Queue (Acquire: Get Service)			Number of Happy Customer	Number of Unhappy Customer	Service Desk Utilization
		Input	Output	Average Waiting Time			
Original	1	91	88	157.82	53	34	100
Experiment 1	2	94	91	47.39	89	0	83
Experiment 2	3	84	84	5.58	82	0	51.33

Service Desk Utilization = Resource: Service Desk Utilization

Resource Utilization (%) =

$$\frac{\text{Total Resource Count} - \text{Average Available Resources}}{\text{Total Resource Count}} \times 100$$

2. Service Time Experiment

*To modify the variability of staff efficiency (Service Baseline), click Delay: Service Time. Go to Delay Properties. Click the drop-down button beside Delay Time. Select: Statistical Distribution → Log Normal 2. Change the Location. Location = 90 is the estimation of staff working 90 seconds for every service.

Experiment	Service Baseline	Average Waiting Time (Acquire: Get Service)	Number of Happy Customer	Service Desk Utilization
Original	90 s	157.82	53	100
Faster Worker	70 s	113.92	64	94
Slower Worker	110 s	166.74	44	100

3. Waiting Tolerance Experiment

*To modify the variability of Max Wait Timer: click Acquire: Get Service. Go to Properties. Make sure Max Wait Timer is ticked. Change the Set Time.

Experiment	Maximum Wait	Average Queue Size (Acquire: Get Service)	Maximum Queue Size (Acquire: Get Service)	Completed Services (No. of Happy Customers)	No. of Unhappy Customers
Original	200 s	2.81	8	53	34
Experiment 1	100 s	1.52	5	52	46
Experiment 2	150 s	2.02	8	53	35
Experiment 3	300 s	4.26	10	53	31

TASK 4: ANALYSIS

Experiment 1: Number of Service Counters Experiment

- 1. How does increasing the number of service counters affect average waiting time and queue performance? Explain why this happens.**

Increasing the service counters drastically reduces the average waiting time from 157.82 seconds down to 5.58 seconds and completely eliminates all unhappy customers. This happens because adding resources expands the parallel processing capacity of the desk, breaking the bottleneck created when customer arrivals outpace the speed of a single staff member.

- 2. Compare Service Desk Utilization for Resource Count = 1, 2, and 3. Which configuration provides the best balance between customer waiting time and resource usage?**

Utilization drops from 100% with one counter, to 83% with two counters, and down to an inefficient 51.33% with three counters. A resource count of two provides the best balance because it keeps staff highly productive and cost-effective at 83% utilization while giving great service by eliminating all unhappy customers and keeping wait times low.

- 3. At which Resource Count does adding more service counters provide little improvement? Explain your reasoning using simulation results.**

Adding a third service counter provides very little practical improvement. While upgrading to two counters successfully eliminates all 34 unhappy customers, pushing the count to three only shaves off a few additional seconds of wait time but causes desk utilization to plunge from 83% to 51.33%, making the extra cost unjustifiable.

Experiment 2: Service Time Experiment

1. **How does changing service duration influence average customer waiting time? Explain the relationship.**

There is a direct relationship where increasing the service duration causes customer waiting times to rise significantly. When service times are shorter (70 seconds), the worker processes people quickly, which minimizes the backlog and keeps the average wait at 113.92 seconds. Conversely, longer service times (110 seconds) keep customers at the counter for too long, causing incoming arrivals to pile up in the queue and pushing the average waiting time up to 166.74 seconds.

2. **Compare the number of happy customers for Faster Worker and Slower Worker scenarios. Why do the results differ?**

The Faster Worker scenario results in a much higher number of happy customers (64) compared to the Slower Worker scenario, which only yields 44. These results differ because a faster service rate shortens the queue wait time, allowing more customers to reach the counter before they lose patience. In contrast, the slower worker creates a stagnant queue, forcing more customers to exceed their maximum waiting tolerance and leave the system unhappy.

3. **How does service efficiency affect Service Desk Utilization? Explain why utilization changes when service duration changes.**

Higher service efficiency reduces utilization, while lower efficiency pushes utilization to its absolute limit. When the worker is fast (70 seconds), they clear the queue so effectively that they create small windows of idle time, dropping utilization to 94%. When service duration increases to 110 seconds, efficiency drops, meaning the worker takes so long per customer that a constant backlog forms, forcing the counter to run at 100% utilization without any downtime.

Experiment 3 : Waiting Tolerance Experiment

- 1. How does increasing customer waiting tolerance affect the number of unhappy customers? Explain the trend observed.**

Increasing the customer waiting tolerance directly reduces the number of unhappy customers, dropping them from 46 at a 100-second maximum wait down to 31 at a 300-second maximum wait. This trend occurs because giving customers a higher patience threshold means they are willing to stand in the queue longer before abandoning it. This extra patience buys the single service counter more time to process the backlog, allowing more people to get served instead of walking away frustrated.

- 2. Compare Average Queue Size and Maximum Queue Size across experiments. Why does changing waiting tolerance influence queue behaviour?**

As waiting tolerance increases from 100s to 300s, the average queue size grows from 1.52 to 4.26, and the maximum queue size expands from 5 to 10. Changing the tolerance influences queue behavior because when customers have low patience, they abandon the line quickly, artificially keeping the physical queue small. When patience increases, fewer people leave prematurely, causing more tokens to accumulate and sit in the queue simultaneously, which naturally inflates both the average and peak queue sizes.

- 3. Which Maximum Wait setting provides the best balance between completed services and customer satisfaction? Support your answer using simulation data.**

The 300-second Maximum Wait setting provides the best operational balance. At this setting, the system achieves the highest customer satisfaction with the lowest number of unhappy customers (31 compared to 46 in Experiment 1), while maintaining peak productivity at 53 completed services. While it does result in a larger average queue size (4.26), it maximizes the service counter's potential to clear the line successfully without letting customers abandon the system too early.